

Superconducting Single-Photon Detectors

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Abstract—Superconducting single-photon detectors are cryogenically cooled superconducting devices enabling the detection of individual photons from the ultraviolet to the mid-infrared spectrum. These are pivotal tools for advancing the frontier of quantum optics and low-light sensing. Among these superconducting detectors, superconducting nanostripe single-photon detectors (SNSPDs or SSPDs) are widely recognized as a leading technology, due to their near-unity detection efficiency, picosecond timing jitter, and negligible afterpulsing. Although they require operation at cryogenic temperatures, their exceptional performance metrics—including high detection efficiency (>90%), low dark count rates, and intrinsic photon-number resolution are indispensable for advancing quantum communication, quantum computation, and deep-space optical communications.

Index Terms—single-photon detectors, detection efficiency, superconducting devices, quantum optics, cryogenics.

I. INTRODUCTION

Superconducting single-photon detectors (SSPDs) are superconducting devices designed to detect the fundamental quanta of light, known as photons, at the individual level. Most implemented as superconducting nanostripe single-photon detectors (SNSPDs), these devices operate just below their critical temperature, acting as ultra-sensitive photon counters. SSPDs are typically operated at cryogenic temperatures (e.g., 0.8–4.2 K) and provide exceptional performance for light waves from the visible to the mid-infrared spectrum [4].

In 2001 [1], the first NbN superconducting single-photon detector was demonstrated by using a single NbN stripe at 4.2 K and current-biased close to its critical current. The device exhibited a measured quantum efficiency of 20% for 0.81 μm wavelength photons and negligible dark counts.

Over the years, various superconducting materials, such as NbN, NbTiN, and WSi, have been implemented for SNSPDs to optimize performance across different wavelength ranges and operating conditions [2]. The highest-performance devices are based on meandering nanostripes (typically 50–200 nm wide and a few nanometers thick) fabricated from ultra-thin superconducting films on dielectric substrates, integrated into commercial cryostat systems. These devices offer single-photon sensitivity from the ultraviolet (<400 nm) to the mid-infrared (>2000 nm) and can achieve system detection efficiencies exceeding 90% at telecommunications wavelengths around 1550 nm [2], [3]. This performance surpasses that of semiconductor-based detectors in key metrics, establishing

SSPDs as the enabling technology for the most demanding quantum optical applications.

II. OPERATIONAL PRINCIPLES OF SSPDs

SNSPDs operate as a current-biased superconducting nanostripe, typically a few nanometers thick and ~ 100 nm wide, cooled well below its critical temperature (T_c) [2]. (i) A direct current (I_{bias}) is applied just below the stripe's critical current (I_c). In this superconducting state, current flows without resistance via Cooper pairs, bound electron pairs with opposite spin and momentum that form a macroscopic quantum ground state [2], [5].

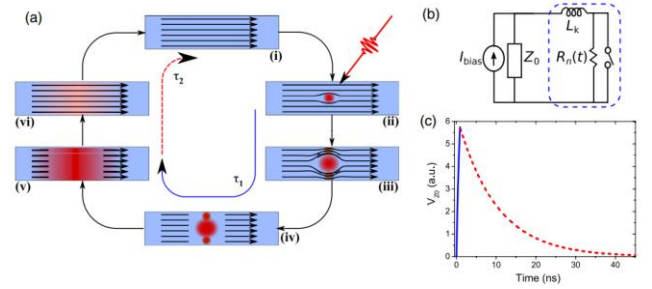


Fig. 1. a) A schematic illustrating the detection cycle of the superconducting nanostripe single-photon detector (SNSPD). b) An electrical equivalent circuit of the SNSPD. c) A simulation of the output voltage pulse of the SNSPD (approximating the pulse shape typically observed on an oscilloscope after amplification). Values of $L_k = 500$ nH and $R_n = 500 \Omega$ (assumed fixed) were used for this simulation [2].

When a photon with energy greater than the superconducting bandgap (2Δ) is absorbed (ii), it breaks Cooper pairs via electron-electron interaction, generating a population of excited single electrons known as quasiparticles. (iii) These quasiparticles rapidly thermalize with the ionic lattice through electron-phonon interactions, emitting phonons and creating a localized, non-superconducting “hotspot” where the superconducting order parameter is suppressed [1], [4]. An instability of the superconducting state emerges due to the quasiparticle cloud, linked with a local reduction of the superconducting order parameter, re-distribution of the current density and lowering of the effective critical current density. Combining ideas from deterministic and stochastic models allows to describe a complete set of measurements qualitatively but these existing models require further developments to be able to fully describe the physical processes quantitatively [9]. (iv) This constriction leads to a local current density that exceeds I_c , triggering a phase transition from the superconducting to the resistive state

across the entire width of the nanostripe [2]. Subsequently, the resistive region of the nanostripe grows due to internal Joule heating.

This sudden appearance of resistance in the order of several kilo Ohms, (v) modeled as a time-varying hotspot resistor $R_{hs}(t)$ in parallel with the stripe's large kinetic inductance L_k , rapidly diverts the bias current into the external readout circuit [9]. The circuit behavior can be described using lumped element models in Fig 1.(b). The kinetic inductance of the device, which dominates due to the inertial mass of the Cooper pairs as the film thickness is much lower than the superconductor London penetration depth, governs the pulse rise time [9]. The resulting voltage pulse, typically a few millivolts after amplification, marks a photon detection event. (vi) The current is then shunted until the hotspot cools via phonon emission and subsequent phonon absorption by the substrate, allowing Cooper pairs to re-form and the nanostripe to recover superconductivity [2]. In cases where the resistive domain does not cool down rapidly enough, the detector latches due to thermal runaway and no further photons can be detected until the device is actively reset.

III. CHARACTERISTICS OF SSPDs

Photon detection efficiency (PDE) of an SSPD is the key parameter describing the probability that a photon incident on the detector system produces a measurable output pulse [2]. To register a detection event, several sequential conditions must be met: the photon must be coupled into the device's active area (coupling efficiency), be absorbed within the superconducting nanostripe (absorption efficiency), and trigger a resistive hotspot that successfully forms a detectable voltage pulse (triggering or registration efficiency) [2]. The absorption efficiency depends critically on the material's intrinsic optical properties and the nanostripe geometry, which is often optimized using optical cavities to enhance absorption at target wavelengths [4]. The triggering probability increases with the bias current closer to the critical current I_c , as a higher supercurrent density makes it easier for a photon-induced hotspot to exceed the critical threshold and create a resistive barrier across the nanostripe [1].

Dark count rate (DCR) is a critical noise parameter representing the average rate of output pulses registered when the detector is shielded from all external light. In SSPDs, dark counts originate primarily from thermal fluctuations (i.e., the spontaneous generation of quasiparticles via phonon absorption) or quantum tunneling of Cooper pairs across the superconducting gap [2], [7]. These mechanisms exhibit strong temperature dependence, which is why SSPDs are operated at deep cryogenic temperatures (typically 0.8–4.2 K) to suppress thermal excitations. Reducing the operating temperature exponentially decreases the DCR, enabling rates as low as a few counts per second [3], [7]. Afterpulsing is virtually negligible in SSPDs as there are no deep-level carrier traps in SSPDs [2].

Recovery time (RT) represents the minimum time between two consecutive photon detections and sets the maximum achievable count rate. In an SSPD, the recovery time is dictated by the time required for the resistive hotspot to

subside and for the bias current to be restored through the nanostripe via its large kinetic inductance L_k . This recovery time is governed by the electron-phonon interaction time and the L_k/R time constant of the readout circuit, where R is the load impedance [2], [5]. For standard nanostripe, dead times are typically in the range of tens of nanoseconds, enabling count rates on the order of tens of MHz.

Timing jitter (TJ) is the timing uncertainty between the arrival of a photon and the generation of the corresponding output pulse. SSPDs exhibit exceptionally low timing jitter, often in the range of a few picoseconds to ~ 100 ps. This jitter arises from the stochastic processes of the initial hotspot formation, its growth dynamics, and electronic noise in the readout amplifier [2], [5]. The intrinsic jitter is linked to the timescale of the nonequilibrium process from Cooper pair breaking to the establishment of a measurable resistance. The histogram of jitter typically has a Gaussian-type distribution and its full-width-at-half-maximum (FWHM) can be defined as the timing jitter of a device.

Photon number resolution (PNR) refers to a detector's ability to distinguish the number of photons in an incident pulse. A standard, single-nanostripe SSPD acts as a binary detector (0 or ≥ 1 photons) because a single photon creates a resistive barrier that diverts the entire bias current. However, advanced SSPD architectures, such as series or parallel arrays of nanostripes (multi-pixel detectors) or tapered nanostripes where the amplitude of the output pulse scales with the number of activated pixels, can provide intrinsic PNR capability [6], [8]. This makes them uniquely valuable for quantum optics and quantum information processing applications that require true photon-number discrimination.

IV. PHOTON NUMBER RESOLUTION FOR SSPDs

Photon-number resolution (PNR) is a critical capability for quantum information protocols such as linear optical quantum computing and boson sampling, which require discrimination between different numbers of photons in a single optical pulse. While conventional single-pixel superconducting nanostripe single-photon detectors (SNSPDs) act as binary devices, advanced configurations enable the retrieval of PNR by analyzing the electrical output signal [6].

One primary method for achieving PNR involves multi-pixel SNSPD architectures. These can be implemented as discrete pixel arrays, where each pixel is an independent nanostripe meander connected to a separate readout line. In such a configuration, the number of activated pixels corresponds directly to the number of absorbed photons, provided the photon flux is spatially spread across the array. More integrated approaches include series-connected nanostripe arrays or superconducting microstripe striplines. In these designs, the absorption of multiple photons creates multiple spatially separated hotspots within a single, continuous superconducting structure. The key metric is that the amplitude or integrated charge of the resulting output voltage pulse scales approximately linearly with the number of activated regions or hotspots. This amplitude discrimination allows the electrical signal to be analyzed to infer the incident photon number [6].

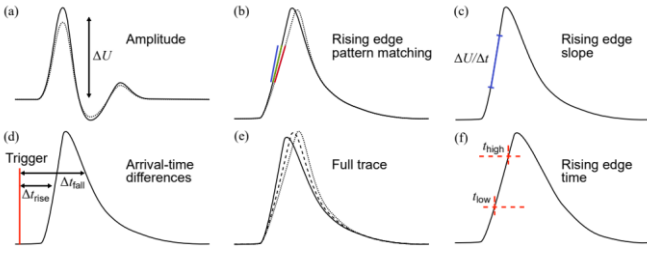


Fig. 2. Schematic representation of techniques to extract photon-number information from SNSPD output signals. (a) Signal amplitude discrimination using an inductor-resistor differentiating circuit or an impedance-matching taper. (b) Waveform analysis via reference rising-edge matching. (c) Linear fitting of the rising-edge slope. (d) Timing discrimination using arrival-time differences relative to a trigger. (e) Principal component analysis (PCA) of a large set of electrical traces. (f) Timing measurement at two distinct threshold levels using a time-to-digital converter [6].

Another promising technique utilizes tapered nanostripe geometries, where the width of the superconducting strip varies linearly. In this design, a photon absorbed in the wider section diverts less of the total bias current than one absorbed in the narrower, more constricted section. Consequently, the amplitude of the output pulse depends on the location of the photon absorption event along the taper. For a multi-photon pulse, the total signal is a superposition of contributions from each absorption site, creating a unique pulse amplitude distribution that can be statistically analyzed to resolve photon numbers. This method intrinsically encodes spatial information into the signal amplitude without requiring physically separated pixels [8]. Optimization of these designs focuses on engineering the nanostripe's kinetic inductance gradient and hotspot dynamics to maximize the separation between pulse amplitude distributions for different photon numbers, thereby improving the fidelity of PNR [6].

V. APPLICATIONS

A. Optical Communications

In deep-space and long-haul terrestrial optical communications, where photon flux is extremely low, SSPDs provide unparalleled sensitivity for reliable signal detection. Their broadband spectral response enables the use of different superconducting materials optimized for key telecommunications wavelengths. Niobium-based (NbN, NbTiN) SSPDs are the dominant platform for the near-infrared, offering high system detection efficiency ($>80\%$), low dark counts, and picosecond timing at the critical 1550 nm and 1310 nm telecom bands [2], [3]. Operation at cryogenic temperatures ($\sim 2-4$ K) is required but manageable in deployed systems. Tungsten silicide (WSi) SSPDs offer performance at longer wavelengths (up to ~ 2000 nm) and demonstrate even lower dark count rates due to a lower superconducting gap, making them suitable for extended-wavelength division multiplexing [4].

B. Quantum Technologies

SSPDs are foundational for modern quantum optics and quantum information science. Their near-unity efficiency, negligible afterpulsing, and excellent timing jitter make them

ideal for detecting entangled photon pairs, performing time-correlated single-photon counting, and enabling long-distance quantum key distribution (QKD) [2], [3]. In fiber-based QKD, the high efficiency and low noise of SSPDs at 1550 nm have been critical in breaking the secret key rate and distance records, enabling practical secure communication over hundreds of kilometers. Their fast reset time also supports high-speed QKD protocols, which are essential for the deployment of quantum networks.

C. Other Major Implementations

The exceptional performance of SSPDs enables breakthroughs in a wide range of scientific and technological fields. Key applications include time-resolved fluorescence spectroscopy in biophysics and chemistry; lidar and laser ranging for precision metrology; optical time-domain reflectometry for fiber network diagnostics; single-molecule detection and microscopy; astrophotonics for faint-light astronomy; and tests of fundamental quantum mechanics [2], [5]. Their ability to precisely timestamp individual photons makes them indispensable tools wherever the ultimate in optical sensitivity and timing precision is required.

VI. CONCLUSION

SSPDs represent a leading technology for quantum and low-light applications, offering near-unity efficiency, picosecond timing, and photon-number resolution. Their exceptional performance, despite requiring cryogenic operation, enables breakthroughs in quantum communications, sensing, and computing. Future advances will focus on improving system integration and expanding operational capabilities.

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